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Started	Wednesday, 21 October 2026, 9:31 PM
Completed	Wednesday, 21 October 2026, 9:34 PM
Duration	3 mins 41 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element
b. The lower middle element is chosen as the middle element ✗
c. The search stops
d. The middle element is chosen randomly

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Linear search ✓
b. Hashing
c. Binary search
d. Dynamic search

Question **3**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search ✗
b. Linear search
c. Interpolation search
d. Binary search

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(1)
c. O👎 ✓
d. O(n log n)

Question **5**
Correct
Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the element is found in the list ✓
- c. When the list contains duplicate elements
- d. When the search algorithm finishes

Question **6**
Correct
Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Hashing
- b. Linear search ✓
- c. Jump search
- d. Binary search

Question **7**
Correct
Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Deleting elements from a list
- b. Determining whether an element is present in a list ✓
- c. Sorting elements in a list
- d. Inserting elements into a list

Question **8**
Correct
Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search is more efficient than binary search on large lists.
- b. Linear search divides the list into halves.
- c. Linear search can be applied to both sorted and unsorted lists. ✓
- d. Linear search requires the list to be sorted.

Question **9**
Correct
Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the left sublist
- b. In the right sublist ✓
- c. At the beginning of the list
- d. In the middle of the list

Question **10**
Correct
Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection
b. Insertion
c. Complexity
d. Bubble ✓

Question **11**
Correct
Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 500
c. 100
d. 1000

Question **12**
Correct
Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist
b. The list must be sorted
c. Number values should only be present
Number values should only be present
d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **13**
Correct
Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list
b. Sort the list
c. Compare the target element with the middle element in the list ✓
d. Divide the list into two equal parts

Question **14**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. n ✓
- b. log n
- c. n/2
- d. 1

Question **15**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 99 ✓
- b. 89 and 94
- c. 90 and 100
- d. 94 and 99